Experiment No. 3

Experiment Title	To perform Exploratory Data Analysis and visualization using Python
Course Outcomes	CO3: Perform exploratory analysis of the data to be used for mining.
Date of Preparation	
Date of Submission	
Signature & Grade	

1. Aim

To perform Exploratory Data Analysis (EDA) on an airline passenger dataset using Python (pandas, Matplotlib and Seaborn) in Google Colab, and to visualize its distributions, relationships and patterns before any data mining is done.

2. Theory

Exploratory Data Analysis is the process of examining a dataset, using summary statistics and charts, to understand its structure, find patterns and unusual values, and decide which attributes are useful for mining. It is done before building any model, so that the model is based on a clear understanding of the data. EDA is carried out at three levels:

- Univariate analysis: looking at one column at a time (its distribution, spread and skewness).
- Bivariate analysis: looking at the relationship between two columns, for example a column against the target.
- Multivariate analysis: looking at three or more columns together.

Chart	What it shows	Used in this experiment for
Bar / count plot	How many records fall in each category	Target and categorical columns
Histogram	Shape and skewness of a numeric column	Age, Flight Distance, delays
Box plot	Median, spread and extreme values, by group	Age and Flight Distance by satisfaction
Grouped bar chart	Averages compared between groups	Service ratings by satisfaction
Correlation bar chart	Strength of the linear relationship	Every numeric column against satisfaction
Heatmap	A value for each pair of categories	Satisfaction rate by Class and Type of Travel

3. Dataset and Tools

Dataset: Airline Passenger Satisfaction (Kaggle), file train.csv, with 103,904 passengers and 25 columns: passenger details (Gender, Customer Type, Age), flight details (Type of Travel, Class, Flight Distance, delays), 14 service ratings from 0 to 5, and the target column 'satisfaction' ('satisfied' or 'neutral or dissatisfied').

Source: kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

Tools: Python 3 in Google Colab with the libraries pandas, NumPy, Matplotlib and Seaborn.

4. Implementation

4.1 Load and Prepare the Data

The columns 'Unnamed: 0' (a row index) and 'id' (an identifier) were dropped because they do not describe the passenger. A helper column 'satisfied_flag' (1 = satisfied, 0 = not) was added so that satisfaction rates and correlations can be calculated. The original data was not changed in any other way.

```
df = pd.read_csv('train.csv')
df = df.drop(columns=['Unnamed: 0', 'id'])
df['satisfied_flag'] = (df['satisfaction'] == 'satisfied').astype(int)
```

4.2 Summary Statistics

```
df.describe().T                      # numeric columns
df.describe(exclude='number').T      # text columns
```

Numeric columns:

	count	mean	std	min	25%	50%	75%	max
Age	103904.0	39.38	15.11	7.0	27.0	40.0	51.0	85.0
Flight Distance	103904.0	1189.45	997.15	31.0	414.0	843.0	1743.0	4983.0
Inflight wifi service	103904.0	2.73	1.33	0.0	2.0	3.0	4.0	5.0
Departure/Arrival time convenient	103904.0	3.06	1.53	0.0	2.0	3.0	4.0	5.0
Ease of Online booking	103904.0	2.76	1.40	0.0	2.0	3.0	4.0	5.0
Gate location	103904.0	2.98	1.28	0.0	2.0	3.0	4.0	5.0
Food and drink	103904.0	3.20	1.33	0.0	2.0	3.0	4.0	5.0
Online boarding	103904.0	3.25	1.35	0.0	2.0	3.0	4.0	5.0
Seat comfort	103904.0	3.44	1.32	0.0	2.0	4.0	5.0	5.0
Inflight entertainment	103904.0	3.36	1.33	0.0	2.0	4.0	4.0	5.0
On-board service	103904.0	3.38	1.29	0.0	2.0	4.0	4.0	5.0
Leg room service	103904.0	3.35	1.32	0.0	2.0	4.0	4.0	5.0
Baggage handling	103904.0	3.63	1.18	1.0	3.0	4.0	5.0	5.0
Checkin service	103904.0	3.30	1.27	0.0	3.0	3.0	4.0	5.0
Inflight service	103904.0	3.64	1.18	0.0	3.0	4.0	5.0	5.0
Cleanliness	103904.0	3.29	1.31	0.0	2.0	3.0	4.0	5.0
Departure Delay in Minutes	103904.0	14.82	38.23	0.0	0.0	0.0	12.0	1592.0
Arrival Delay in Minutes	103594.0	15.18	38.70	0.0	0.0	0.0	13.0	1584.0

Text columns:

	count	unique	top	freq
Gender	103904	2	Female	52727
Customer Type	103904	2	Loyal Customer	84923
Type of Travel	103904	2	Business travel	71655
Class	103904	3	Business	49665
satisfaction	103904	2	neutral or dissatisfied	58879

Figure 1: Summary statistics of the numeric and text columns

The data has 103,904 passengers. Ages range from 7 to 85 with a mean of 39.4. Flight Distance ranges from 31 to 4,983 with a median of 843 and a mean of 1,189, so a few long flights pull the mean above the median. Both delay columns have a median of 0 but maximums of 1,592 and 1,584 minutes, which shows a very long tail. The service ratings have means between 2.73 and 3.64. Arrival Delay has 103,594 values, so 310 are missing.

4.3 Target Variable, Missing Values and Duplicates

```
df['satisfaction'].value_counts()
df.isnull().sum()
df.duplicated().sum()
```

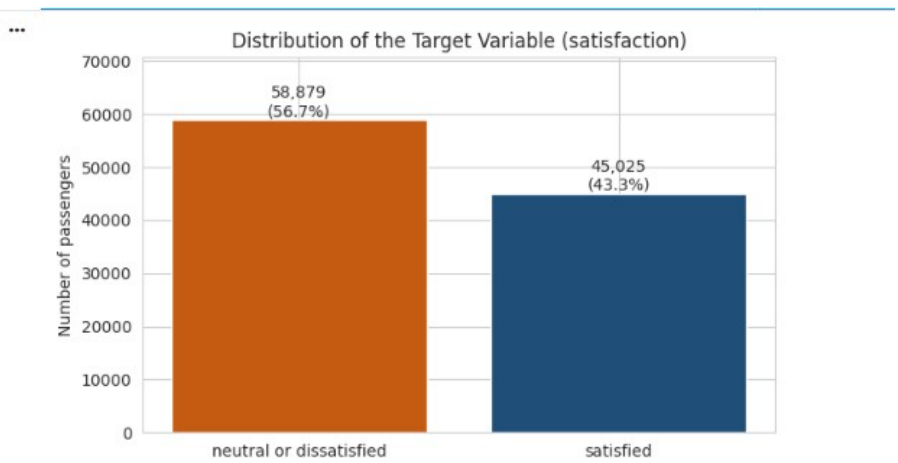


Figure 2: Distribution of the target variable, missing values and duplicates

56.7% of the passengers (58,879) are neutral or dissatisfied and 43.3% (45,025) are satisfied, so the two classes are reasonably balanced. Only Arrival Delay in Minutes has missing values (310, about 0.3%), and there are no duplicate rows.

4.4 Univariate Analysis: Categorical Columns

```
sns.countplot(data=df, x='Class', order=df['Class'].value_counts().index)
```



Figure 3: Number of passengers in each category

Men and women are almost equally represented (51,177 and 52,727). Most passengers are loyal customers (84,923 against 18,981), most travel for business (71,655 against 32,249), and almost all fly Business (49,665) or Eco (46,745) class. Eco Plus is small (7,494).

4.5 Univariate Analysis: Numeric Columns

```
sns.histplot(df['Age'], bins=40, kde=True)
df['Age'].skew()
```

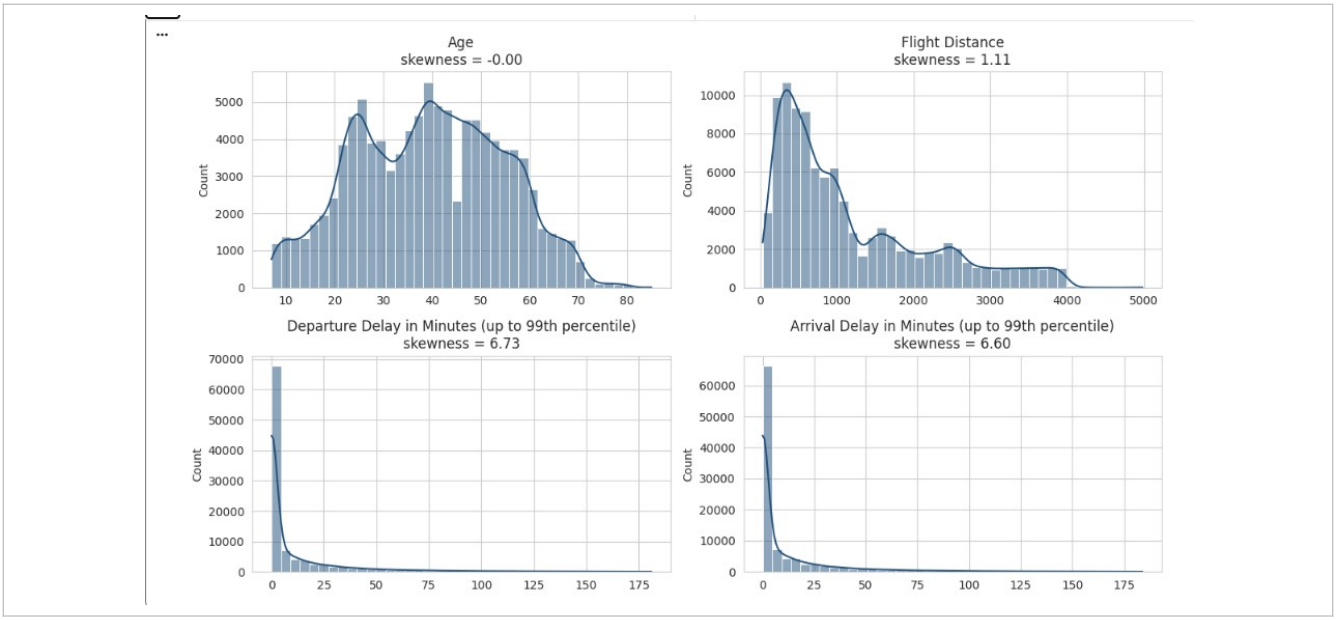


Figure 4: Histograms of Age, Flight Distance and the two delay columns

Age is symmetric (skewness about 0) with peaks near 25 and 40. Flight Distance is right-skewed (skewness 1.11): most flights are under about 1,000 and a smaller group flies 1,500 to 4,000. The delay columns are extremely right-skewed (6.73 and 6.60): most passengers have no delay or a very small one, and a few have very long delays. The delay histograms show values only up to the 99th percentile so the shape stays readable.

4.6 Bivariate Analysis: Columns against Satisfaction

```
sns.boxplot(data=df, x='satisfaction', y='Age')
(df.groupby('Class')['satisfied_flag'].mean() * 100)
```

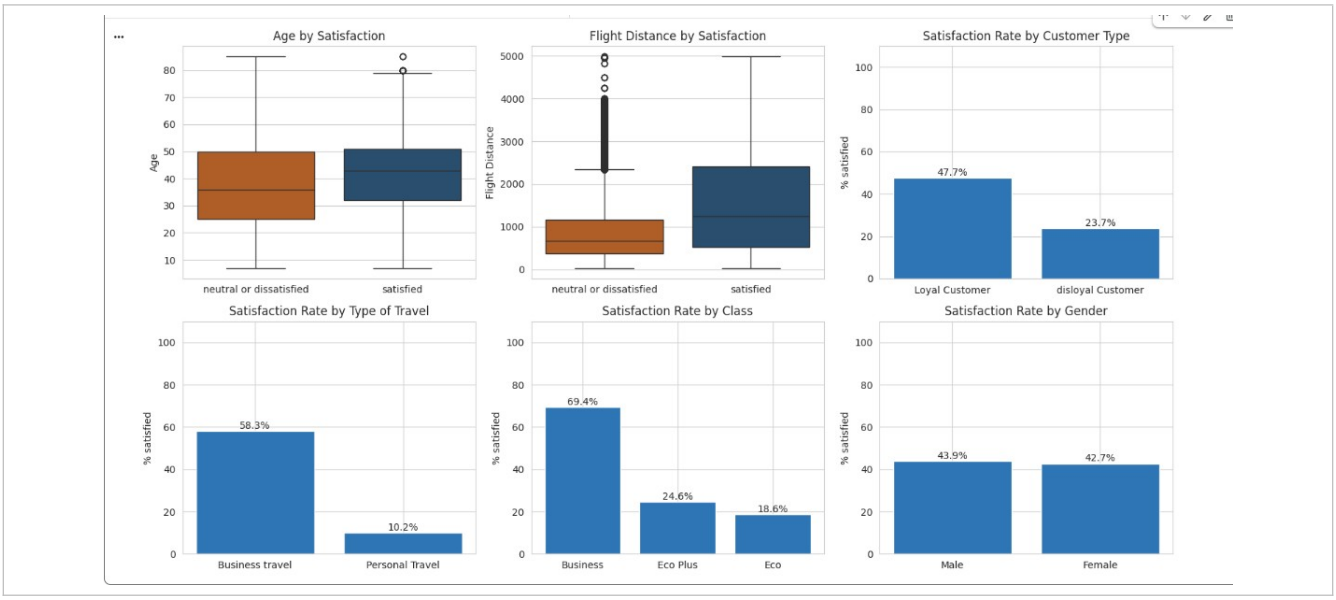


Figure 5: Age and Flight Distance by satisfaction, and satisfaction rate for each category

Satisfied passengers are older and fly longer distances than the neutral or dissatisfied group. The satisfaction rate is much higher for loyal customers (47.7% against 23.7%), for business travel (58.3% against 10.2%) and for

Business class (69.4%, against 24.6% for Eco Plus and 18.6% for Eco). Gender makes almost no difference (43.9% for men and 42.7% for women).

4.7 Service Ratings

```
df.groupby('satisfaction')[rating_cols].mean().T.plot(kind='barh')
```

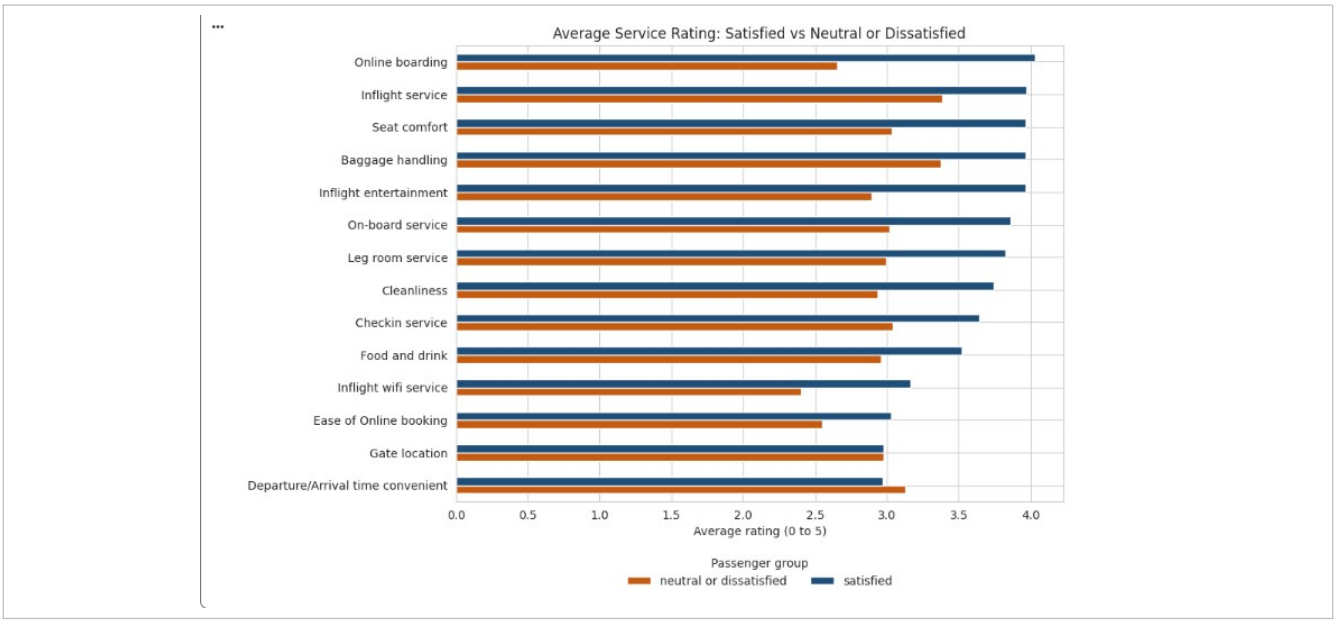


Figure 6: Average rating of each service for satisfied and neutral or dissatisfied passengers

Satisfied passengers give higher ratings to almost every service. The biggest gaps are for Online boarding (about 4.0 against 2.7), Inflight entertainment and Seat comfort. Gate location is rated about the same by both groups, and Departure/Arrival time convenient is rated slightly higher by the neutral or dissatisfied group.

4.8 Correlation with Satisfaction

```
df.select_dtypes(include='number').corr()['satisfied_flag'].sort_values()
```

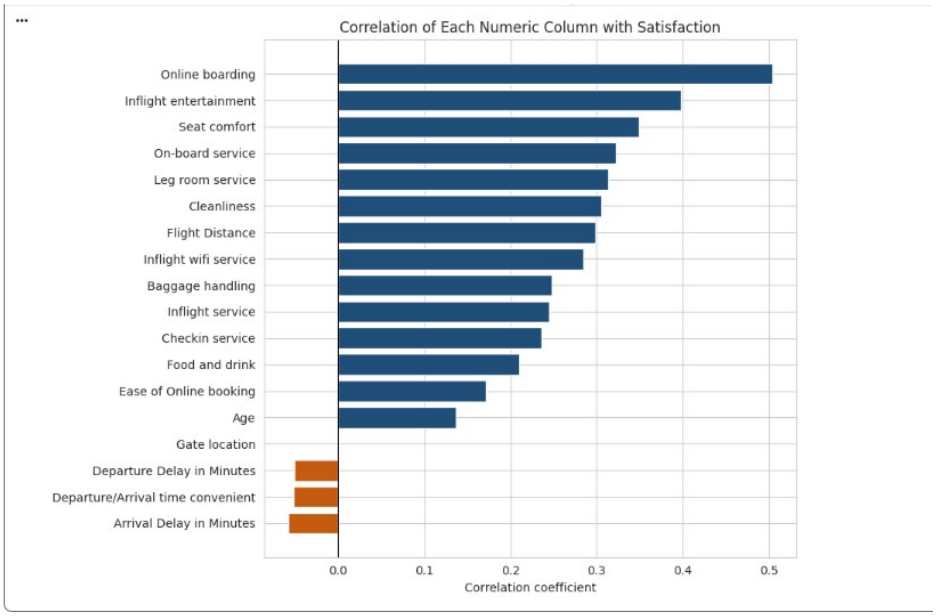


Figure 7: Correlation of each numeric column with satisfaction

Online boarding has the strongest relationship with satisfaction (0.50), followed by Inflight entertainment (0.40) and Seat comfort (0.35). Flight Distance and several service ratings are around 0.3. Gate location has almost no relationship, and Departure Delay, Arrival Delay and Departure/Arrival time convenient are very slightly negative (−0.05 to −0.06).

4.9 Multivariate Analysis: Class and Type of Travel

```
pivot = df.pivot_table(values='satisfied_flag', index='Class',
                       columns='Type of Travel', aggfunc='mean') * 100
sns.heatmap(pivot, annot=True, fmt='.1f', cmap='Blues')
```

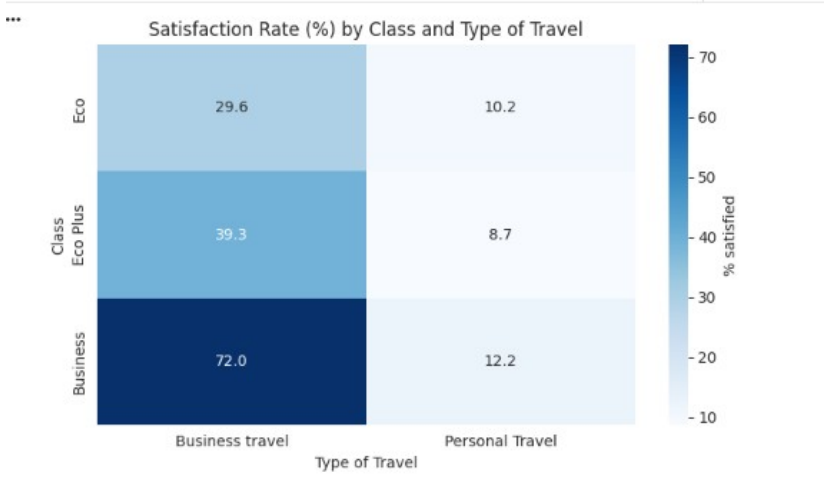


Figure 8: Satisfaction rate (%) by Class and Type of Travel

On business trips the satisfaction rate rises sharply with class: 29.6% in Eco, 39.3% in Eco Plus and 72.0% in Business. On personal trips it stays very low in every class (8.7% to 12.2%). So the purpose of the trip matters, and class makes a big difference only for business travellers.

4.10 Key Findings

```
Overall satisfaction rate: 43.3%

Class          : Business 69.4%,  Eco Plus 24.6%,  Eco 18.6%
Type of Travel : Business travel 58.3%,  Personal Travel 10.2%
Customer Type  : Loyal Customer 47.7%,  disloyal Customer 23.7%
Gender         : Male 43.9%,  Female 42.7%

Strongest positive correlation with satisfaction:
   Online boarding: 0.50
   Inflight entertainment: 0.40
   Seat comfort: 0.35
Strongest negative correlation with satisfaction:
   Departure Delay in Minutes: -0.05
   Departure/Arrival time convenient: -0.05
   Arrival Delay in Minutes: -0.06

Skewness of Departure Delay in Minutes: 6.73
Skewness of Arrival Delay in Minutes: 6.60
```

Figure 9: Key findings printed from the data

5. Summary of Findings

Finding	Evidence
Overall satisfaction	43.3% of passengers are satisfied; the classes are reasonably balanced
Data quality	Only 310 missing values (Arrival Delay in Minutes); no duplicate rows
Most related ratings	Online boarding (0.50), Inflight entertainment (0.40), Seat comfort (0.35)
Type of Travel	Business travel 58.3% satisfied; personal travel only 10.2%
Class	Business 69.4%, Eco Plus 24.6%, Eco 18.6%
Best group	Business class on business trips: 72.0% satisfied
Customer Type	Loyal 47.7% against disloyal 23.7%
Gender	No real difference (43.9% against 42.7%)
Delays	Extremely right-skewed (6.73 and 6.60) and only very weakly related to satisfaction

6. Conclusion

Exploratory Data Analysis and visualization were performed on the Airline Passenger Satisfaction dataset using Python in Google Colab. Summary statistics showed the structure of the data and that only 310 Arrival Delay values are missing, with no duplicates. Univariate charts showed that Age is symmetric, Flight Distance is moderately right-skewed and the delay columns are extremely right-skewed. Bivariate and multivariate charts showed that satisfaction depends most on Online boarding, Inflight entertainment, Seat comfort, the type of travel and the travel class, and hardly at all on gender or delays.

These findings show which attributes are most useful for mining. They agree with the Decision Tree in Experiment 4, which also split first on Online boarding and Type of Travel. The delay columns will need outlier treatment and the missing Arrival Delay values will need to be filled before mining, as done in the preprocessing experiment.